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(54) **BAG CLOSURE SYSTEM**

(71) Applicant: **Christopher Telman**, Franklin, TN (US)

(72) Inventor: **Christopher Telman**, Franklin, TN (US)

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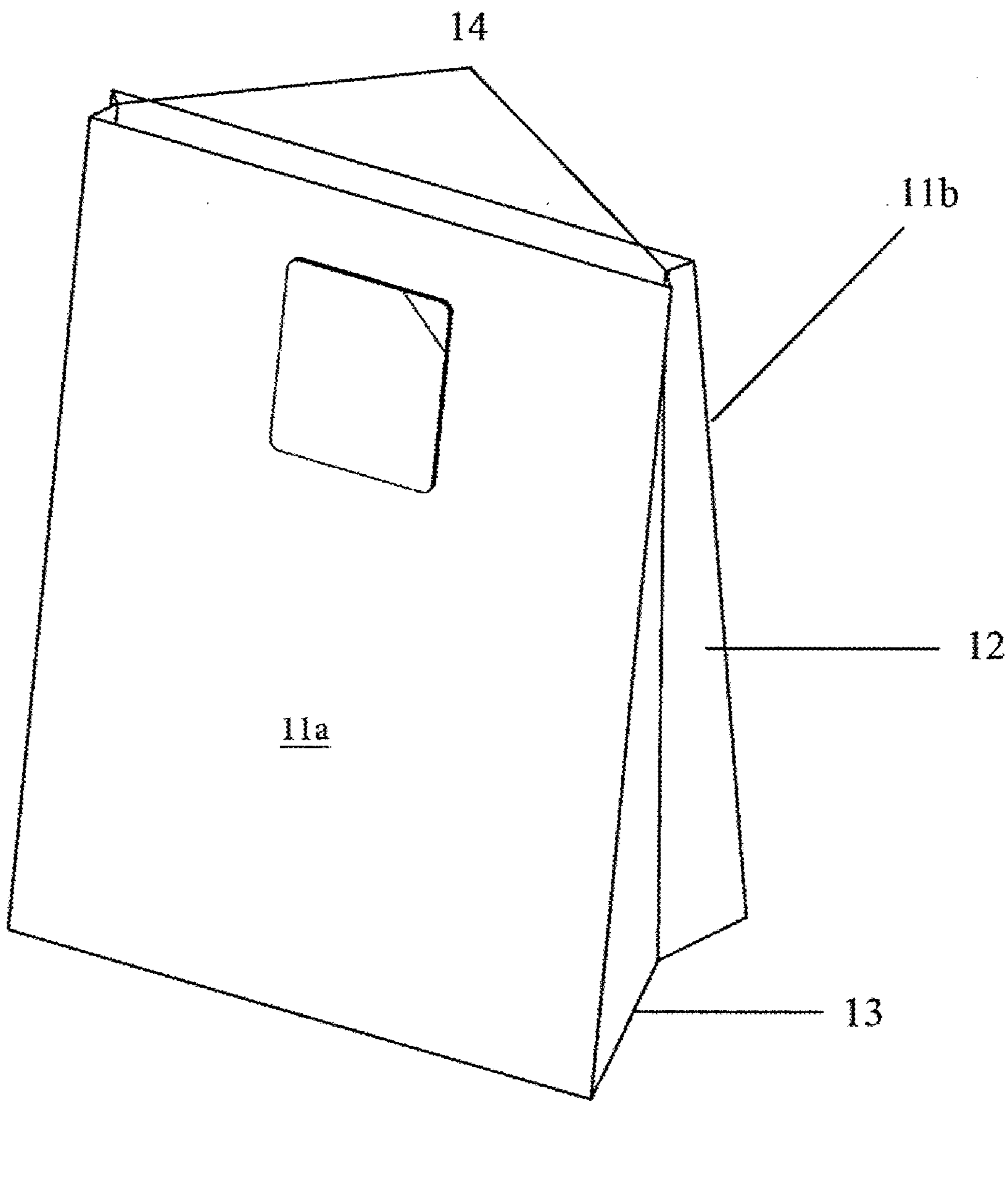
CPC *B65D 77/14* (2013.01)

(57) **ABSTRACT**

A system and method for sealing a flexible container such as a paper-based bag having two sets of opposing panels that can be pressed together and folded over each other. The system includes a strip of material having an adhesive on a single side of a top ply that is attached to a surface of a second or bottom ply that is partially coated with a release coating such that that part of the top ply is releasably attached to the bottom ply and part of it is permanently attached to the bottom ply. The user removes the portion of the top ply from the portion of the bottom ply that is coated with a release coating, partially folds the top edges panels and places the bottom surface of the top ply over opposite panels of the bag. Some embodiments have a second top ply attached to the first top ply.



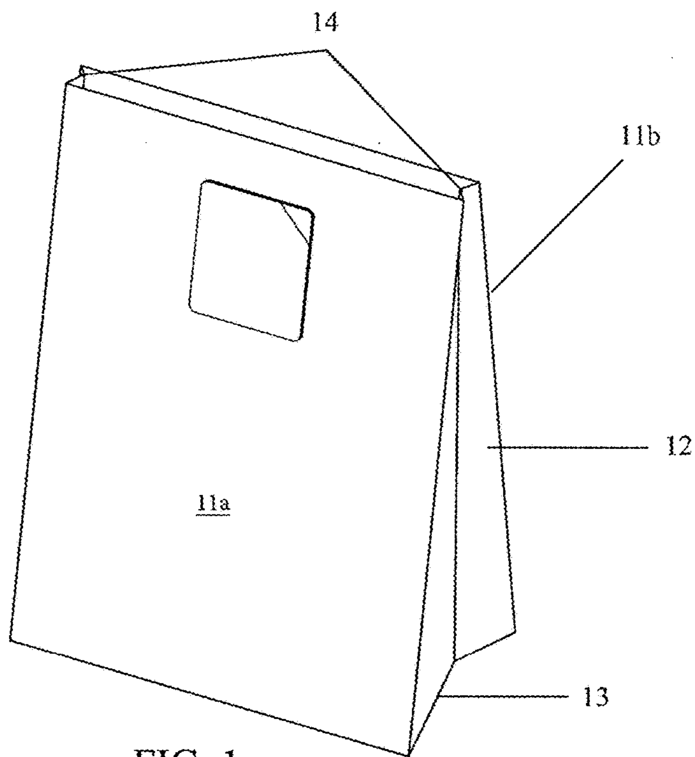


FIG. 1

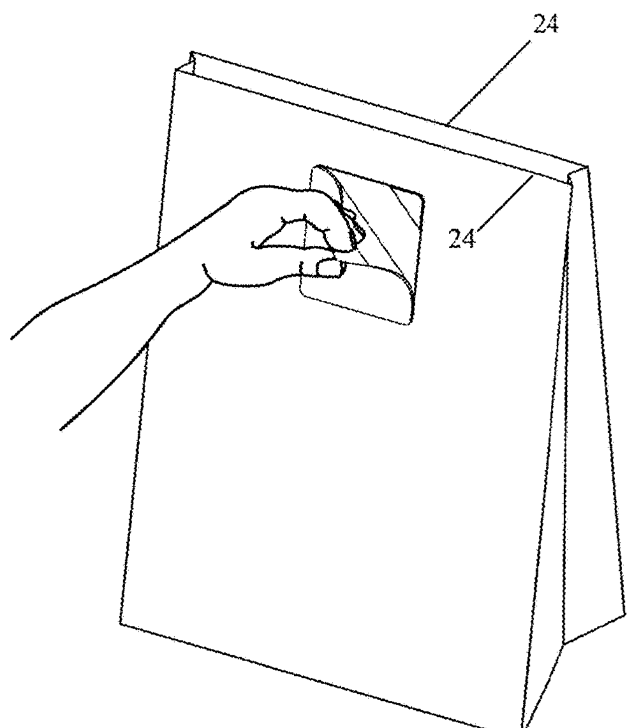


FIG. 2

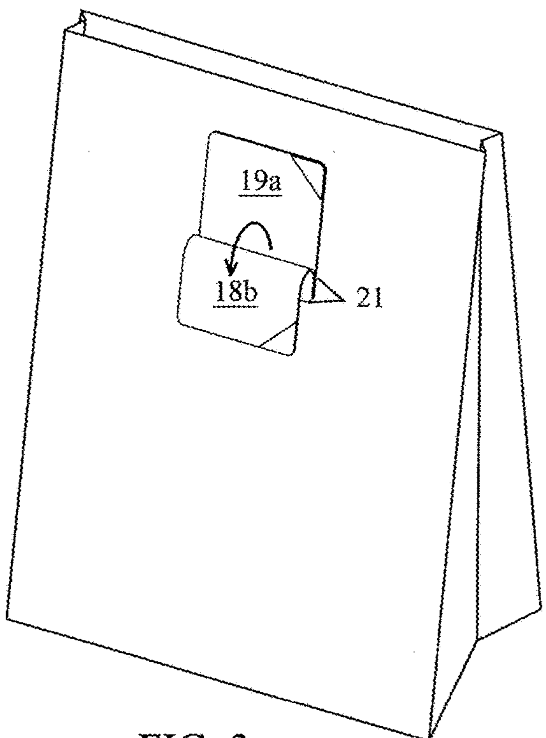


FIG. 3

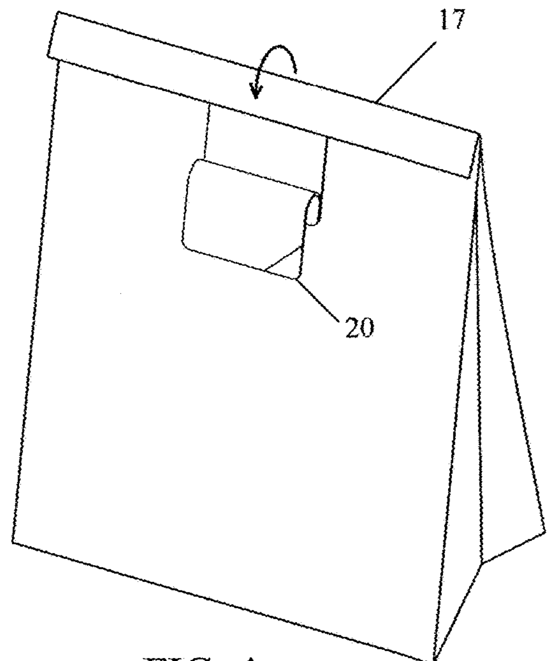


FIG. 4

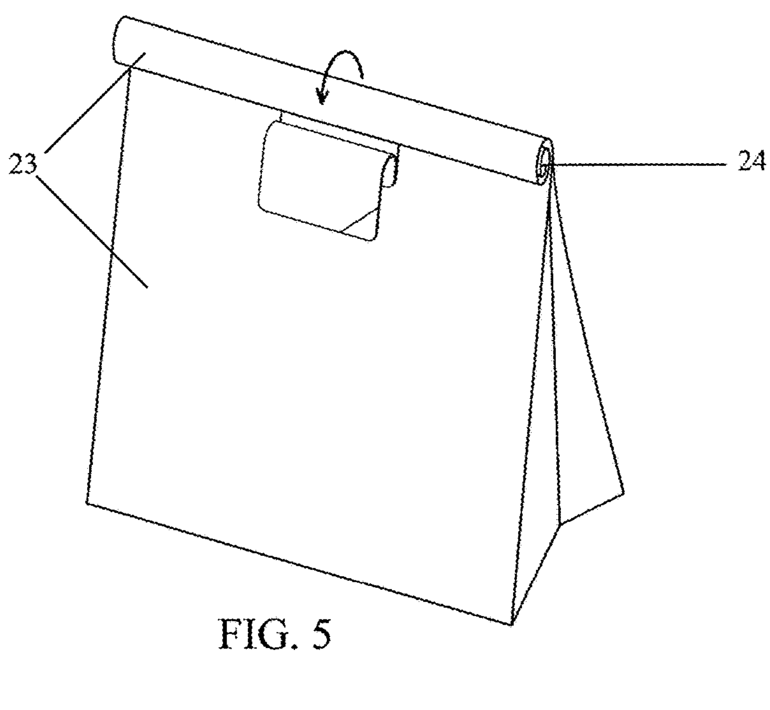


FIG. 5

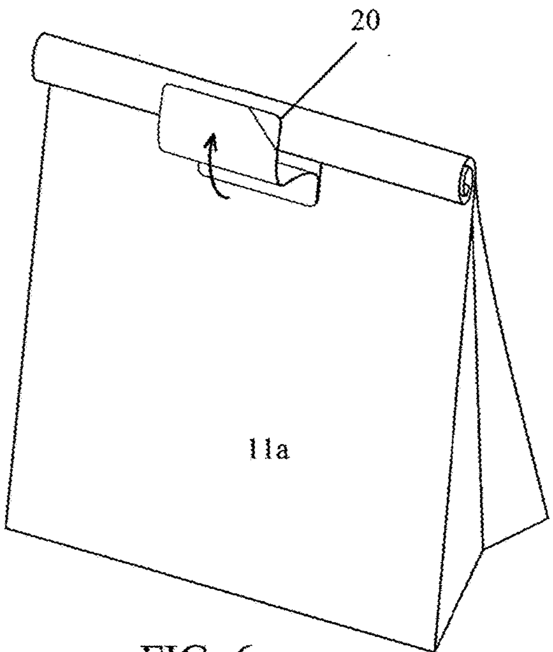


FIG. 6

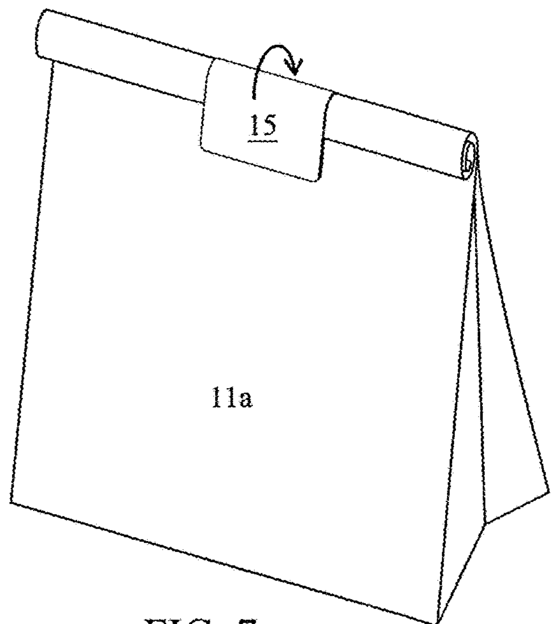


FIG. 7

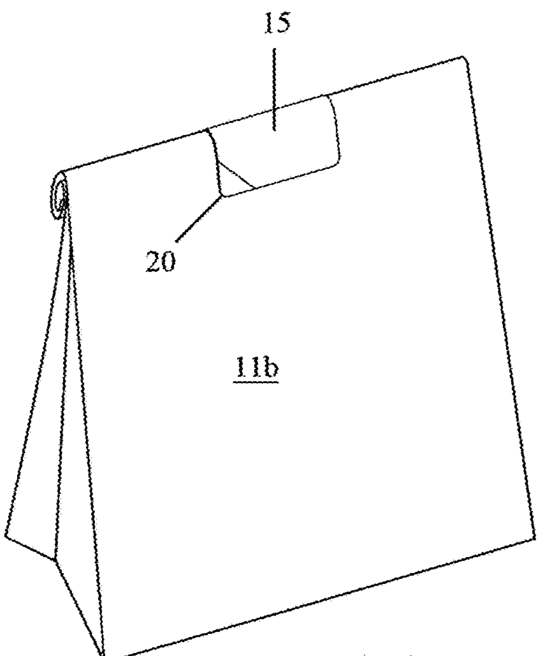


FIG. 8

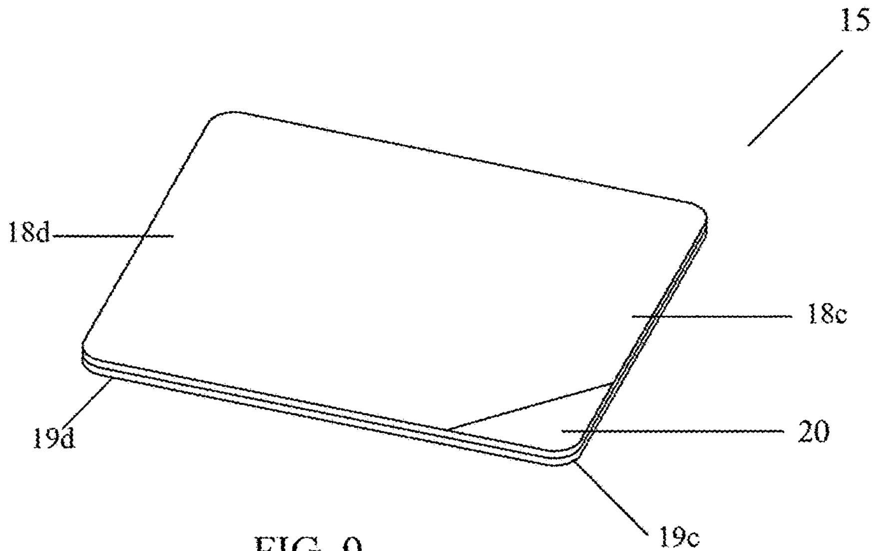


FIG. 9

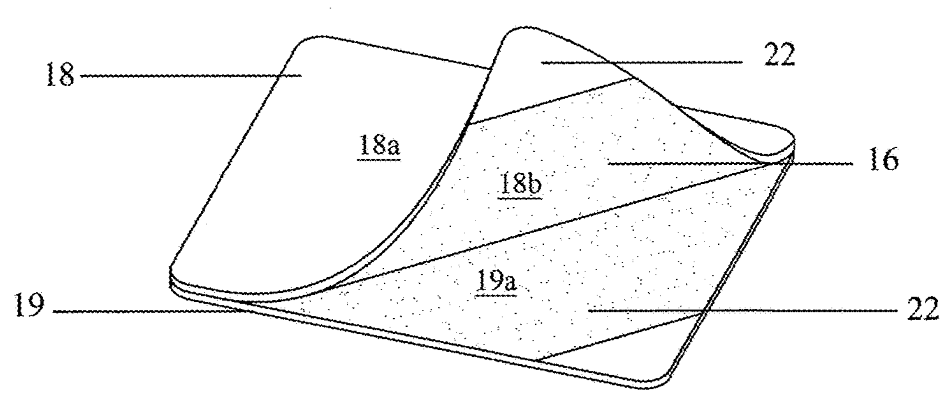


FIG. 10

BAG CLOSURE SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to U.S. Provisional Application No. 63/561,083 filed on Mar. 4, 2024. The content of U.S. Provisional Application No. 63/561,083 filed on Mar. 4, 2024 is incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

(a) Field of the Invention

[0002] The present invention is in the technical field of packaging. More particularly, the present disclosure relates to a novel method of sealing shipping bags.

[0003] In this disclosure the term “features” means that the stated structure is a prominent feature or aspect of another stated structure, i.e. the featured structure is included in, attached to or otherwise directly associated with the other stated structure.

BRIEF SUMMARY OF THE INVENTION

[0004] The present invention is an apparatus and method for sealing a bag, such as, but not limited to a common paper bag used to enclose, transport and/or ship consumer goods. For illustrative purposes, the disclosure will focus on a standard paper bag with a closed bottom and an opposing open top with four sides enclosing a cavity or chamber in which items are placed for temporary storage. These bags typically have two sets of opposing sides and are temporarily closed by folding two opposing sides in their approximate center so that the other two opposing sides of the bag physically align such and their interior surfaces are at least partially flattened together and are in contact with each other. A skilled artisan will appreciate upon reviewing the following disclosure, that this system can be used to seal virtually any container that has flexible sides and an opening at at least one end. In addition, references to front, back and side panels are for convenience only. This disclosure focuses on a method of sealing a bag using two opposing side panels regardless of whether they are front and aback or side panels.

[0005] Preferred embodiments of the container or bag closure system include a strip or length of adhesive material such as a sticker or tape. Preferred embodiments and the anticipated best mode of practicing this invention include a strip of material that has two layers, i.e a top ply and a bottom ply. The top ply is attached to the bottom ply and the bottom ply is attached directly to an outside surface of the container or bag, proximate to the open end of the bag. The bottom ply has a first or bottom surface and a second or top surface. The bottom surface includes an adhesive coating that attaches the first/bottom surface of the bottom ply to the bag or container. The top or second surface of the bottom ply is partially coated with a release coating, such as but not limited to a silicone coating. The release coating is applied to that top or second surface except for a portion of that top surface that is farthest away from the opening of the bag. This second portion or section of the top surface of the bottom ply does not feature any adhesive or release coating. The top ply also has a first or bottom surface and a second or top surface. The first/bottom surface of the top ply is coated with an adhesive and is applied directly to the top surface of the bottom ply. Because of the partial release coating on the to surface of the bottom ply, the first or bottom surface of the top ply adheres permanently to the second portion or section of the top surface of the bottom ply, but is releasably attached to the rest of the top surface of the bottom ply. The strip of adhesive, including the top and bottom ply, are positioned near the open end of the bag.

[0006] In use, the user can peel most of the top ply away from the bottom ply leaving a lower portion (second end) of the top ply attached to the bottom ply. Doing so reveals the adhesive side of the top ply such that the user can then fold the ends of the panels of the bag to close the open end of the bag. When doing so, the strip of adhesive (the top ply) is positioned to be laid over the folded portion of the bag allowing the adhesive to adhere to the outside surface of the folded portion of the bag thereby holding the folded sides of the bag together and sealing the open end of the bag. The strip of adhesive, when so placed adheres to the outside surface of two opposing sides of the bag thereby sealing the open end of the bag. When the top ends or edges of two or more sides of a bag are folded over one another, the bag tends not to stay in that configuration, but rather it will tend to return to its originally unfolded state. Placing the strip of material such that it attaches to a panel of the bag, over the folded portion of the bag and onto the opposing panel of the bag prevents the fold from unfolding. This configuration prevents shearing on both sides of the bag, i.e. it prevents the label from being torn away from the bag to which it is adhered.

[0007] Other embodiments of the bag sealing system include a tab or a portion of the bottom surface of the top ply of the strip of adhesive material that is not covered by an adhesive coating. In preferred embodiments, a release coating is applied to a corner of the bottom surface of the top ply. Ideally, this corner is near the top of the strip of adhesive material, i.e. it is positioned on the portion of the top play that is closest to the opening of the bag. In use, the user bends the open ends of the sides of the bag away from the side to which the strip of material containing adhesive is attached. This causes the portion of the top ply featuring the tab to physically separate from the bottom ply allowing the user to grab the tab and pull the top ply partially away from the top surface of the bottom ply. The user then folds the edges of the sides of the bag adjacent to the open end of the bag over on themselves one or more times. At that point, the user simply lays the bottom surface of the top ply that is detached from the bottom ply over the outside or exterior surface of the folded panels of the bag resulting in the strip of adhesive material being connected to the exterior surfaces of two opposing panels thereby holding the two opposing sides of the bag together.

[0008] Some embodiments of the bag closure system include a crease in the bag that facilitates folding the opening of the bag. Pre-folding the ends of two of the sides of the portion of the bag that flanks or encircles the open end of the bag creates a crease that makes the final closure of the bag require less effort.

[0009] Still other embodiments of the bag closure system/method include more than one strip of adhesive material as described above and below being positioned on the same or opposing exterior surfaces of the bag’s panels. Still other embodiments of the system include one or more of the strips of adhesive material being attached to an inside surface of one or two opposing panels of the bag. The strips of adhesive material described herein can be manufactured on a roll and attached to and peeled away from a release liner during manufacture.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0010] FIG. **1** is a front perspective view of bag featuring the bag closure system prior to use;

[0011] FIG. **2** is a front perspective view thereof showing a user grasping the tab and partially peeling the top ply away from the bottom ply;

[0012] FIG. **3** is a front perspective view thereof showing the top ply peeled all the way back to the hinge portion while the bottom ply stays attached directly to the bag;

[0013] FIG. **4** is a front perspective view thereof showing the folding of the open ends of the bag;

[0014] FIG. **5** is an additional front perspective view thereof;

[0015] FIG. **6** is a front perspective view thereof showing the top ply being placed over the exterior surfaces of the bag;

[0016] FIG. **7** is a front perspective view thereof showing the top ply having been placed over the outer surfaces of the bag; and

[0017] FIG. **8** is a rear perspective view thereof showing the completely sealed bag;

[0018] FIG. **9** is a perspective view of the label itself; and

[0019] FIG. **10** is a perspective view of the top ply partially peeled back from the bottom ply.

DETAILED DESCRIPTION OF THE INVENTION

[0020] FIG. **1** shows a preferred embodiment of the bag closure system. This disclosure uses a typically paper-based bag **10** with two sets of opposing panels-a front **11***a* and back **11***b* panel and the two side panels **12**. These panels are joined to a bottom panel **13** at one end and define an opening or open end **14** at the opposing ends of said panels. The bag can be partially closed by pressing the front panel **11***a* and back panel **11***b* together so that the top edges of the front and back panels **11***a*, **11***b* that are adjacent to the opening **14** are flattened against each. FIG. **1** also shows the strip of material **15**, made up of a top ply **18** and a bottom ply **19**, attached to the bag **10** near the point where the two opposing front and back panels **11***a*, **11***b* have been folded to form a crease **17**. Note, the terms front and back and side are arbitrary and interchangeable. When referring int this disclosure to front and back panels being used to seal the bag, those designations are arbitrary in that any set of opposing panels can be used with this disclosure. The panels need not be the front or back panels, but only need to be opposing panels.

[0021] As shown in FIGS. **9** and **10**, the strip of material **15** includes a top ply **18** and a bottom ply **19** each of which have a top surface **18***a*, **19***a* and a bottom surface **18***b*, **19***b*. A bottom surface **18***b* of the top ply **18** is coated with an adhesive **16** (see FIGS. **9** and **10**). The bottom ply **19** includes a release coating **22** that extends partially along the top surface **19***a* of the bottom ply **19**. This release coating **22** does not extend all of the way from one end **19***c* of the bottom ply **19** to the other **19***d*, but rather the end **19***d* of the bottom ply **19** that is positioned farthest away from the opening **14** does not have the release coating **22** applied to it. Instead, the adhesive **16** that is applied to the bottom surface **18***b* of the top ply **18** non-releasably binds to the top surface **19***b* of the bottom ply **19** at the end **19***d* that does not have a release coating **22** thereby forming a hinge **21**. Note, in the inventor's anticipated best mode, more than half of the bottom ply **19** is coated with a release coating **22** and only a small portion of the top surface **19***a* of the bottom ply **19** is permanently attached to the top ply **18**.

[0022] FIG. **2** also shows the tab **20** which is a portion of the bottom surface **18***b* of the top ply **18**—generally a corner—that is coated with a release coating **22**. Preferred embodiments include the tab **20** being located at a second end **18***d* of the top ply **18** of the strip of material **15**. When the top ply **18** is laid flush against the bottom ply **19**, the release coating **22** on the tab area **2** of the top ply **18** is in contact with the release coating **22** that is present on the top surface **19***c* of the bottom ply **19**. FIG. **2** shows the user grasping the tab **20** and pulling it downward away from an exterior surface **23** of the bag **10**. It is also possible for the user to pull the front and back panels **11***a*, **11***b* downward and away from the strip of material **15** causing the tab **20** to become separated from the bottom ply **19** making it easier for the user to grasp the top ply **18** by the tab **20**.

[0023] FIG. **3** shows the top ply **18** disengaged from the bottom ply **19** at one end, **18***c*, but still attached at an opposing end, **18***d*, to the bottom ply **19**. FIGS. **4** through **7** show the user folding the opposing front and back panels **11***a* and **11***b* of the bag **10** over on themselves to temporarily close the bag **10** and to position the ends of the front and back panels **11***a* and **11***b* of the bag **10** such that the top ply **18** can be placed over the exterior surfaces **23** of both the front panel **11***a* and the back panel **11***b* of the bag **10**. FIG. **8** shows the sealed bag **10** that results from the above process. In this embodiment, the side of the top ply **18** featuring the adhesive **16** is attached to the exterior surfaces of two opposing panels of the bag **10** resulting in a sealed bag.

[0024] FIGS. **9** and **10** show images of the strip of material **15**. Preferred embodiments of the top ply **18** also include a hinge area **21** that is located in the top ply **18** at the point where the top ply **18** contacts the bottom ply **19** where the release coating **22** stops. Some embodiments feature a tab **20** or a portion of the bottom surface **18***b* of the top ply **18** that includes a release coating or otherwise is configured to not to attach to the bottom ply or the bag **10**. This tab **20** makes it easier for the user to grip the top ply **18** and peel it away from the bottom ply **19**.

[0025] Still other embodiments of the bag closure system can include multiple strips of material **15** positioned in different places on the exterior surfaces **23** or the interior surfaces of the bag **10**. Some embodiments include a plurality of strips of material **15** as described above all positioned one or more exterior surfaces **23** of the bag **10** near the opening **14**. While less desirable, the same strips of material **15** can be placed on an interior surface of one or both of the opposing front **11***a* and back panels **11***b*. In fact, these strips of material **15** can also be placed on side panels **12** of the bag **10**. In addition, the strip of material **15** can be positioned to fit over two opposing sides of the bag **10** at a point that is not adjacent to the open end **14** of the bag **10**, but rather the strip of material **15** can be positioned such that top ply **18** extends from a front panel **1** la to a back panel **11***b* without covering the opening **14** of the bag **10**. Further, the sides **12** of the bag **10** can be folded to form a crease or fold **17** that runs at least part of the way from the opening of the bag **10** to the bottom panel of the bag **10**. In the above-described embodiment, that means that the two shorter sides **12** of the bag can be folded such that the fold or crease **17** is perpendicular to the crease **17** formed when the open end **14** of the bag **10** is closed as described above.

[0026] In use, the user bends the top edges **24** of the sides of the bag **10** away from the side to which the strip of material **15** is attached. This causes the tab **20** to physically separate from bottom ply **19** allowing the user to grab the tab **20** and pull the majority of the top ply **18** away from the bottom ply **19** featuring the release coating. The user then folds top the edges **24** of the front and back panels **11***a* and **11***b* of the bag **10** over on themselves one or more times. At that point, the user simply lays the top ply including the adhesive **16** over the exterior surface **23** of the folded sides of the bag **10** causing the adhesive **16** that is already attached to an exterior surface **23** of one panel of the bag **10** to be attached to the exterior surface **23** of the opposing panel of the bag **10** holding the two opposing panels **11***a*, **11***b* of the bag **10** together.

[0027] Other embodiments of the bag closure system **10** include multiple layers of material joined end to end. In other words, a first or top ply **18** is attached to a bottom ply **19** just as described above. An additional or second top ply **18** with a bottom surface coated with an adhesive can be applied to a top surface **18***a* of the first top ply **18**. Said top surface **18***a* of the first top ply **18** can be partially coated with a release coating **22** so that the bottom surface **18***b* of the second top ply **18** can be releasably attached at one end and non-releasably attached at an opposing end to the top surface **18***a* of the first top ply **18**. When the user separates the first top ply **18** from the bottom ply **19** and the second top ply **18** from the top surface **18***a* of the first top ply **18** an accordion-like structure is formed with adhesive exposed on the bottom surfaces of the first and second top ply and that can be laid across the exterior surfaces of two opposing panels of the bag **10**.

[0028] Yet another embodiment of the bag closure system incudes a release coating on a portion of the bottom surface of the bottom ply (the surface that comes into contact with the exterior surface of one of the bag's panels). This configuration leaves the bottom ply partially attached to the exterior surface of the panel to which it is attached, i.e, a portion of the bottom ply's bottom surface stays attached to the panel and a portion can be easily lifted away from that exterior surface. In this way, when the user peels a portion of the top ply away from the bottom ply, a portion of the bottom ply can also be separated from the exterior surface of the panel allowing the user to lift that portion of the bottom ply when applying the top ply to the opposing panel. Preferred embodiments of this configuration include a bottom ply with adhesive coated on the bottom surface of the bottom ply said coating extending from a first end **19***c* of the bottom surface of the bottom ply toward the second end **19***d*, but not reaching the second end **19***d* of the bottom ply. In this manner, the bottom ply stays partially attached to the exterior surface of the panel to which it is attached. When the user lifts the first end of the top ply, the second end of the top ply that is attached by adhesive to the second end of the bottom ply, pulls that second end of the bottom ply away from the exterior surface of the panel to which it is attached and the user can lay the bottom surface of the top ply over the exterior surface of the opposing panel. In this manner, the length of the strip of adhesive material can be extended.

[0029] Each example embodiment disclosed herein has been included to present one or more different features. However, all disclosed example embodiments are designed to work together as part of a single larger system or method. This disclosure explicitly envisions compound embodiments that combine multiple previously-discussed features in different example embodiments into a single system or method.

[0030] Additionally, unless expressly stated to the contrary, the terms 'first', 'second', 'third', etc., are intended to distinguish the particular nouns they modify (e.g., element, condition, node, module, activity, operation, etc.). Unless expressly stated to the contrary, the use of these terms is not intended to indicate any type of order, rank, importance, temporal sequence, or hierarchy of the modified noun. For example, 'first X' and 'second X' are intended to designate two 'X' elements that are not necessarily limited by any order, rank, importance, temporal sequence, or hierarchy of the two elements. Further as referred to herein, 'at least one of and 'one or more of can be represented using the' (s)' nomenclature (e.g., one or more element(s)).

[0031] Reference throughout the specification to features, advantages, or similar language does not imply that all of the features and advantages that may be realized with the present invention should be or are in any single embodiment of the invention. Rather, language referring to the features and advantages is understood to mean that a specific feature, advantage, or characteristic described in connection with an embodiment is included in at least one embodiment of the present invention. Thus, discussion of the features and advantages, and similar language, throughout the specification may, but do not necessarily, refer to the same embodiment.

[0032] Furthermore, the described features, advantages, and characteristics of the invention may be combined in any suitable manner in one or more embodiments. One skilled in the relevant art will recognize that the invention can be practiced without one or more of the specific features or advantages of a particular embodiment. In other instances, additional features and advantages may be recognized in certain embodiments that may not be present in all embodiments of the invention.

[0033] It is understood that the above-described embodiments are only illustrative of the application of the principles of the present invention. The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiment, including the best mode, is to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims, if any, in conjunction with the foregoing description.

[0034] While the foregoing written description of the invention enables one of ordinary skill to make and use what is considered presently to be the best mode thereof, those of ordinary skill will understand and appreciate the existence of variations, combinations, and equivalents of the specific embodiment, method, and examples herein. The invention should therefore not be limited by the above described embodiment, method, and examples, but by all embodiments and methods within the scope and spirit of the invention.

[0035] Clause 1. A method of sealing a bag having a first panel, an opposing second panel and an opening between the first panel and the second panel comprising the steps of applying a strip of adhesive material to an exterior surface of the first panel; separating a portion of the strip of adhesive material away from the exterior surface of the first panel revealing a bottom surface of the strip of adhesive material coated with an adhesive; applying the bottom surface of the strip of adhesive material to an exterior surface of the second opposing panel without removing the strip of adhesive material from the exterior surface of the first panel.

[0036] Clause 2. A method of sealing a bag having a front panel, an opposing back panel and an opening between the front panel and the back panel comprising the steps of forming a strip of material comprised of a top ply with a bottom surface coated with an adhesive; a bottom ply with a top surface coated with a release coating; wherein the bottom surface of the top play is applied directly to and laid flush against the top surface of the bottom ply thereby releasably attaching the top ply to the bottom ply at a first end and non-releasably attaching the top ply to the bottom play at a second end forming a hinge; applying an adhesive to a bottom surface of the bottom ply; applying the bottom surface of the bottom ply to an exterior surface of the front panel at a point that is adjacent to the opening; separating a portion of the top ply away from the bottom ply revealing the bottom surface of the top ply; applying the bottom surface of the top ply to an exterior surface of the back panel.

[0037] Clause 3. The method of Clause 3 further comprising the steps of folding a top edge of the back panel over a top edge of the front panel before applying the bottom surface of the top ply to an exterior surface of the back panels.

[0038] Clause 4. The method of Clause 4 further comprising the steps of applying a release coating to a corner of the bottom surface of the top ply proximate to the first end of the top ply.

[0039] Clause 5. A method of sealing a bag having a front panel, an opposing back panel and an opening between the front panel and the back panel comprising the steps of coating a bottom side of a top ply of a strip of material with an adhesive; partially coating a top surface of a bottom ply of the strip of material with a release coating; attaching the bottom surface of the top ply to the top surface of the bottom ply forming an assembled strip of material wherein a first portion of the top ply is non-releasably attached to the bottom ply and a second portion of the top ply is releasably attached to the bottom ply.

[0040] Clause 6. The method of Clause 6 further comprising the steps of removing the top ply from the portion of the bottom ply that is coated with a release coating; folding a top edge of the back panel over a top edge of the front panel; applying the bottom surface of the top ply that is no longer attached to the bottom ply to an exterior surface of the back panel thereby attaching the front and to the back panel with the top ply.

[0041] Clause 7. The method of Clause 7 further comprising the steps of providing the top ply with a flexible portion that serves as a hinge when the top ply is partially removed from the bottom ply.

[0042] Clause 8. The method of Clause 8 further comprising the steps of applying a release coating to a corner of the bottom surface of the top ply.

I claim:

1. A method of sealing a bag having a first panel, an opposing second panel and an opening between the first panel and the second panel comprising the steps of:

applying a strip of adhesive material to an exterior surface of the first panel;

separating a portion of the strip of adhesive material away from the exterior surface of the first panel revealing a bottom surface of the strip of adhesive material coated with an adhesive;

applying the bottom surface of the strip of adhesive material to an exterior surface of the second opposing panel without removing the strip of adhesive material from the exterior surface of the first panel.

2. A method of sealing a bag having a front panel, an opposing back panel and an opening between the front panel and the back panel comprising the steps of:

forming a strip of material comprised of:

a top ply with a bottom surface coated with an adhesive;

a bottom ply with a top surface coated with a release coating;

wherein the bottom surface of the top play is applied directly to and laid flush against the top surface of the bottom ply thereby releasably attaching the top ply to the bottom ply at a first end and non-releasably attaching the top ply to the bottom play at a second end forming a hinge;

applying an adhesive to a bottom surface of the bottom ply;

applying the bottom surface of the bottom ply to an exterior surface of the front panel at a point that is adjacent to the opening;

separating a portion of the top ply away from the bottom ply revealing the bottom surface of the top ply;

applying the bottom surface of the top ply to an exterior surface of the back panel.

3. The method of claim **3** further comprising the steps of folding a top edge of the back panel over a top edge of the front panel before applying the bottom surface of the top ply to an exterior surface of the back panels.

4. The method of claim **4** further comprising the steps of applying a release coating to a corner of the bottom surface of the top ply proximate to the first end of the top ply.

5. A method of sealing a bag having a front panel, an opposing back panel and an opening between the front panel and the back panel comprising the steps of:

coating a bottom side of a top ply of a strip of material with an adhesive;

partially coating a top surface of a bottom ply of the strip of material with a release coating;

attaching the bottom surface of the top ply to the top surface of the bottom ply forming an assembled strip of material wherein a first portion of the top ply is non-releasably attached to the bottom ply and a second portion of the top ply is releasably attached to the bottom ply.

6. The method of claim **6** further comprising the steps of:

removing the top ply from the portion of the bottom ply that is coated with a release coating;

folding a top edge of the back panel over a top edge of the front panel;

applying the bottom surface of the top ply that is no longer attached to the bottom ply to an exterior surface of the back panel thereby attaching the front and to the back panel with the top ply.

7. The method of claim **7** further comprising the steps of providing the top ply with a flexible portion that serves as a hinge when the top ply is partially removed from the bottom ply.

8. The method of claim **8** further comprising the steps of applying a release coating to a corner of the bottom surface of the top ply.

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